

Fitness Misinformation Detection System

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General Problem

Health misinformation on social media is a significant public health concern. The WHO identifies it as a **critical threat**, as **false claims** reach **vulnerable populations** across **digital platforms**.

Specific Problem

Exercise and nutrition misinformation is particularly concerning, yet **existing fact-checking systems** focus on **political or medical content**.
No **automated systems** verify **fitness-related claims** at scale.

Problem Statement

Digital platforms lack **automated systems** for identifying **exercise and nutrition misinformation**. This enables **harmful wellness advice** to spread, leading to **injuries, unrealistic expectations, and unsafe practices**.

Proposed Solution

An **AI-Powered** system that verifies **exercise and nutrition claims** against **peer-reviewed evidence**. It integrates **NLP** with **audio transcription** for **automated fact-checking** with **confidence scores and citations**.

Trget Audence

User Category	Stakeholders
Primary Users	Fitness enthusiasts and beginners; Individuals making training decisions
Secondary Beneficiaries	Fitness professionals and trainers; Health educators and researchers; Public health organizations

Research Objectives

Objective	Deliverable
Review literature on fitness misinformation detection	Comprehensive understanding of detection approaches
Analyze gaps in automated verification approaches	Identification of system requirements
Curate 500+ peer-reviewed exercise science papers	Evidence database for claim verification
Define categories for claim classification	Taxonomy for exercise/nutrition assertions

System Development Objectives

Component	Goal
NLP System	Claim extraction and classification from text
ML Model	Validity assessment with confidence scores (75%+ accuracy target)
Audio Processing	Integrate OpenAI Whisper API for transcription
Web Interface	User-friendly platform with medical disclaimers
Validation	System accuracy testing with known misinformation examples

Agile Scrum — 6 Sprints over 24 weeks

- Sprint 1: Literature review, database curation, setup
- Sprint 2: NLP pipeline, claim extraction
- Sprint 3: ML model training and validation
- Sprint 4: Web interface, REST API
- Sprint 5: System integration and testing
- Sprint 6: Deployment, bug fixes, documentation

Resources

Category	Technology/Tools
Frontend	React.js
Backend	Python (ML), Node.js (API)
Database	PostgreSQL, Redis (caching)
AI/NLP	OpenAI Whisper API, TensorFlow/PyTorch, Hugging Face
Deployment	AWS/GCP, Docker
Development	8GB RAM minimum, GitHub, VS Code
Research Sources	PubMed, Google Scholar, exercise science journals

Literature Review

Theme	Key Findings	Sources
Misinformation Growth	Exponential increase in health misinfo research post-COVID; 88% of Kenyans encounter online misinformation	Yeung et al. (2022); Wasserman & Madrid-Morales (2019)
Fitness-Specific Gaps	Exercise misinfo mixes scientific jargon with unproven claims ; only 31% of nutrition websites provide evidence-based content	Marocolo et al. (2021); Denniss et al. (2023)
Detection Limitations	Existing systems focus on political/medical content , not fitness; 82-85% accuracy for general medical misinfo, but lack domain specialization	Wang et al. (2020); Schlicht et al. (2024)
Kenyan Context	Rapid social media adoption outpaces digital literacy ; cultural factors influence credibility assessment	Tully (2022); Wasserman & Madrid-Morales (2019)
Technical Approaches	BERT-based models effective for health text; hybrid human-AI systems achieve 78% accuracy but face scalability issues	Upadhyay et al. (2023); Barve & Saini (2021)

Project Gantt Chart

Dec 2025

Jan 2026

Feb 2026

Mar 2026

Tasks

Months

